

ISSUE 01 • VOLUME APEX • 2026-05

The race engineer for the drivers who don't have one.

A professional race engineer costs hundreds of pounds per day. Every Formula 1 driver has one. Most adaptive racers, veteran-team drivers, and grassroots competitors do not. APEX changes that.

BUILT ON IBM GRANITE

ADAPTIVE RACERS FIRST

APEX.RACE

Regulatory barrier dropped in *2017*. Economic barrier stayed.

*T*he FIA lifted its single-seater ban on disabled drivers in December 2017. The barrier stopped being regulatory. It became economic. A paid race engineer is a luxury good for amateur and clubman series.

Three constituencies, one shared product gap: adaptive racers, veteran-team drivers, and grassroots clubman racers. Current AI race-engineer tools (Track Titan, Trophi.ai, Deep Dynamics) assume able-bodied physics. APEX is the first integrated workflow for adaptive hand-controls.

*The
bec*

20

FIA APPENDI

Three files in. *Coaching report* out.

Sarah uploads her telemetry CSV, her FIA Certificate of Adaptations PDF, and a 30-second written debrief at the tensor level. It extracts the hand-control hardware specification and produces the COA-overlap flag downstream.



Telemetry

01

1-Hz mini-sector tensors. Speed, acceleration, lateral G, RPM.

COA PDF

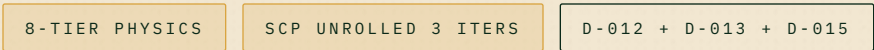
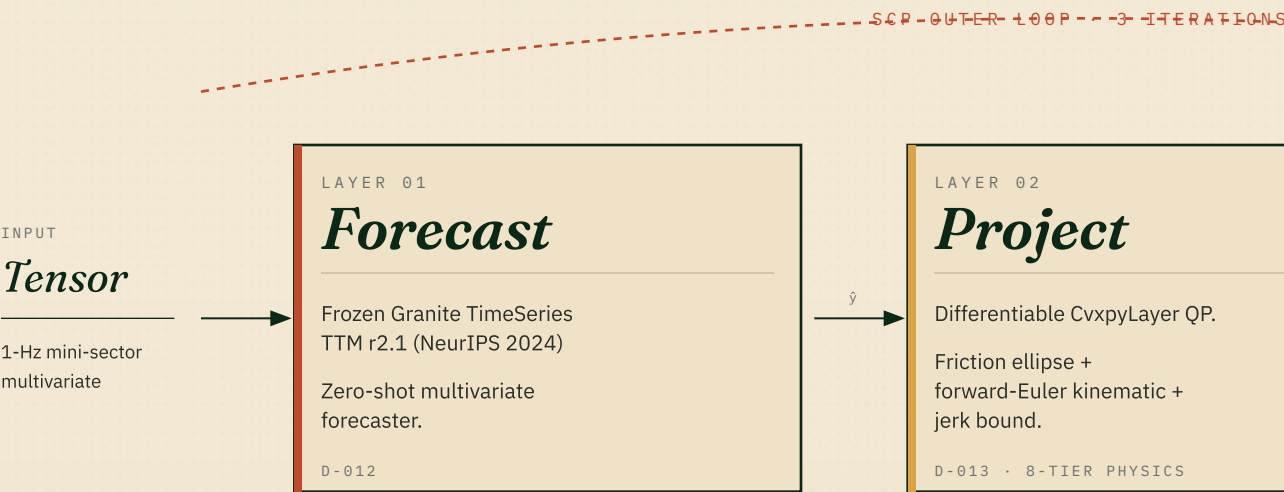
FIA Appendix L Certificate of Adaptations. Granite-Docling tensor parse.

Coaching Report

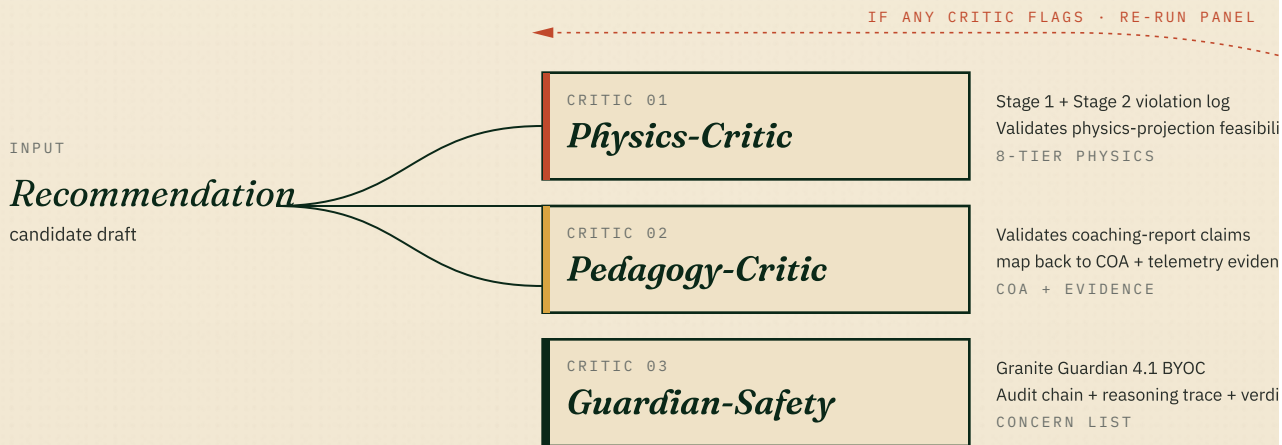
→ OUT

Corner-by-corner insights. Tuning delta. Forecast envelope. Guardian audit. Provenance footer on every claim.

Three layers between *tensor* and *recommendation*.



Three agents read every recommendation *before the driver does*



If any critic flags, IBM Mellea's Instruct-Validate-Repair fires with a loop budget of three until the panel approves. Three independent agents plus the safety gate.

Provenance on every line.

Corner-by-corner insights. Tuning recommendations. Forecast envelope. Every claim cites a specific FIA Appendix L provision and a specific COA section via the provenance footer.

The Watson TTS walkie-talkie reads the report in race-engineer voice with the paddock-radio filter chain (highpass=350, lowpass=3000, compand, volume=1.8). Sarah hears it on her cool-down lap. Web Speech API fallback on environments without FFmpeg (per D-043 Vercel architectural constraint).

FIA APPENDIX L

COA SECTION CITE

WATSON TTS

WALKIE-TALKIE FILTER

HIGHPASS

350 Hz

LOWPASS

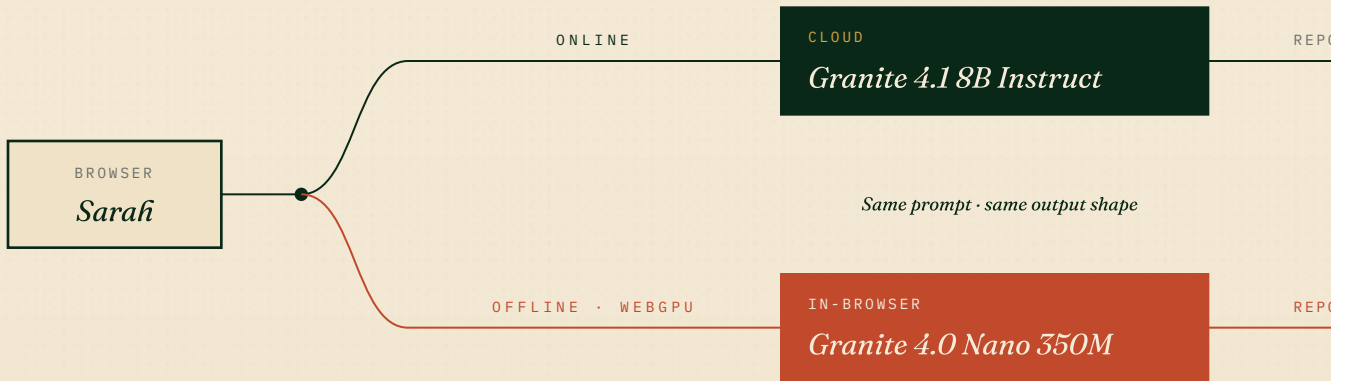
3000 Hz

VOLUME

1.8 ×

Sub-15-second coaching *without a connection.*

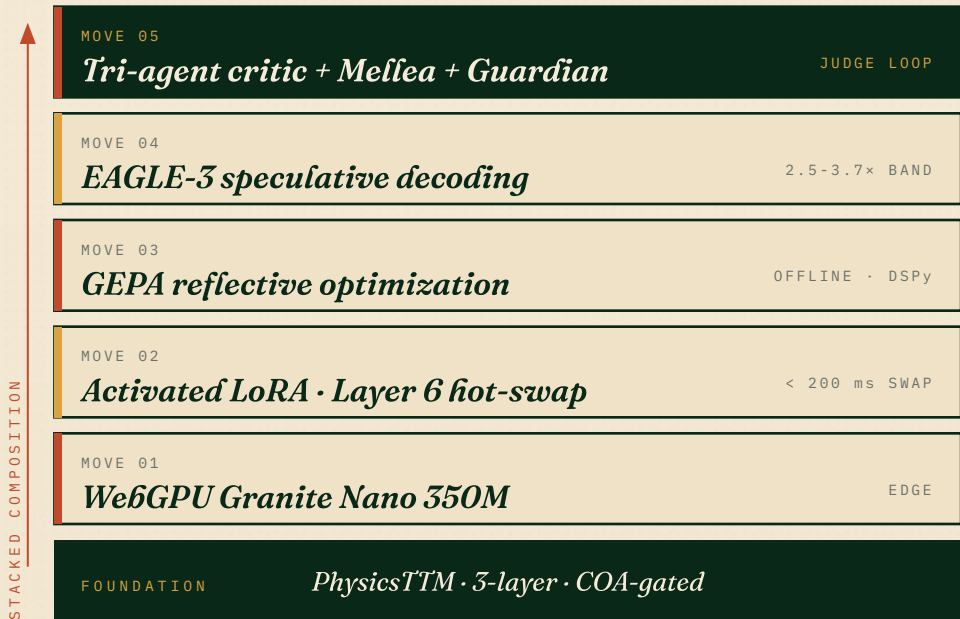
If Sarah loses connection in the paddock, the Granite 4.0 Nano 350M model runs in her browser via WebGPU. Same prompt. Same output shape. No round-trip.



350M



Stacked on top of the physics · critic · audit chain.



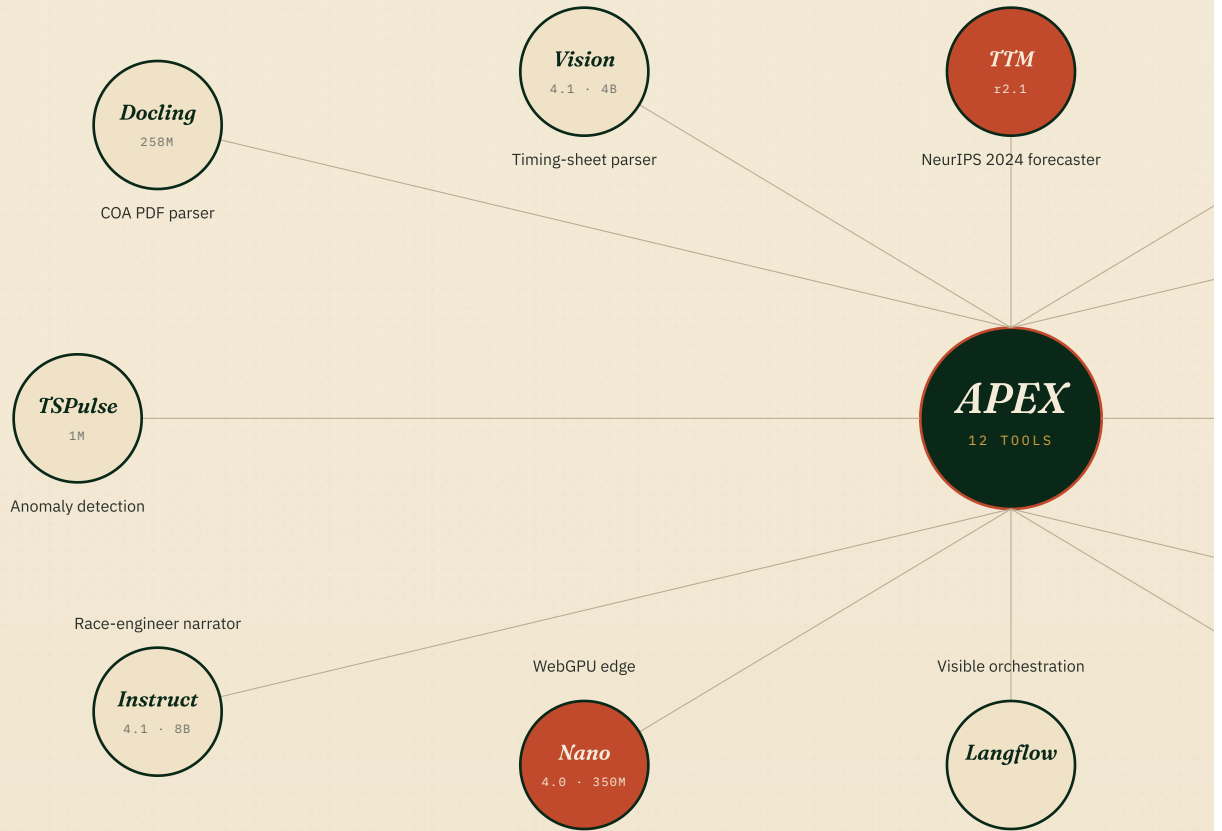
- 01 *We6GPU Granite Nano 350M*
no re
- 02 *Activated LoRA · Layer 6 hot-swap*
pres
- 03 *GEPA reflective optimization*
gene
- 04 *EAGLE-3 speculative decoding*
typic
- 05 *Tri-agent critic + Mellea + Guardian*
+ Gu

COA-parameterized brake-throttle simultaneity gate.

The same coaching pipeline produces different physics corrections for adaptive versus able-bodied drivers, governed by the driver's FIA Certificate of Adaptations.

When the COA permits simultaneous brake + throttle (as adaptive racing programmes and adapted-hand-control systems commonly do), Stage 2's feasibility filter recognises it. When the COA does not permit it, the gate flags the input. To our literature review through 2026-Q2, the first public AI race-engineer workflow we found that reads the COA at the tensor level.

Twelve IBM tools. Every slot *load-bearing*.



CLOSE • BUILT ON IBM GRANITE

For adaptive racers *first*.
Every grassroots competitor *second*.

WHERE TO FIND IT

apex.race

github.com/StephenSook/apex

APACHE 2.0 • PUBLIC DAY 1

APEX.RACE

GITHUB.COM/STEPHENSOOK/APEX

APACHE 2.0